

Analysis of the Obstacle Potential Function

Contents

1	Potential Function	1
1.1	Potential Field Plots	2
1.2	Force Field	2
1.3	Sensitivity to τ_x, τ_y	3
1.4	Velocity at a Fixed Point	3
2	Possible Issues	3
2.1	Anomalous Trough in the Force Field	3
2.2	Sensitivity to τ_x, τ_y and Parallel Forces	4
2.3	Fixed Velocity Behavior	4
2.4	Overparameterization Relative to the Degrees of Freedom	4

1 Potential Function

$$U_{obst} = \frac{\lambda(\Delta v / \Delta v_{max})}{k(\Delta x, \Delta y, \Delta v_x, \Delta v_y)} \quad (1)$$

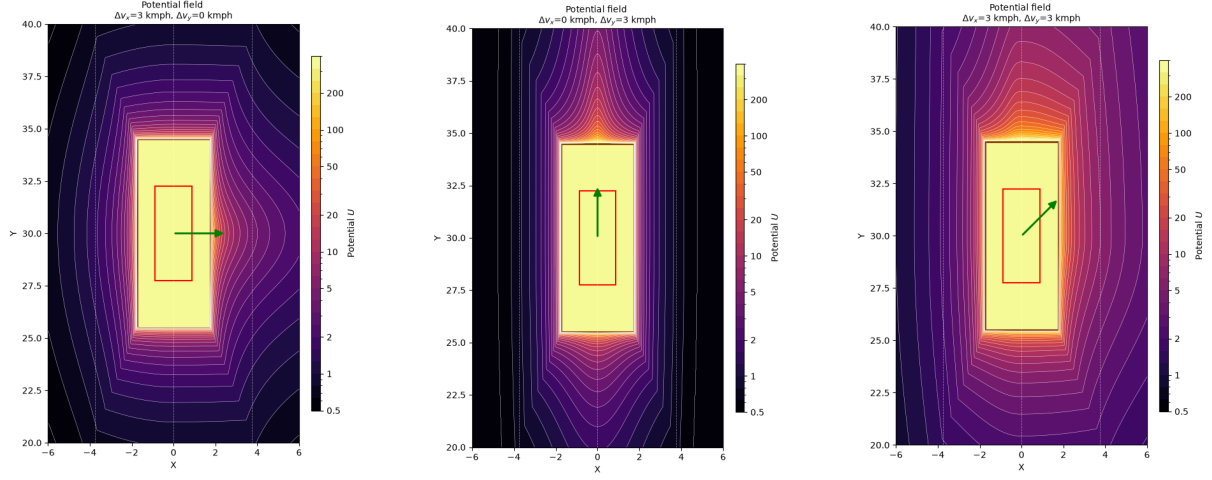
$$k(\Delta x, \Delta y, \Delta v_x, \Delta v_y) = \begin{cases} 0, & \text{if } |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{if } |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{otherwise} \end{cases} \quad (2)$$

$$\tau(\Delta x, \Delta v_x) = \frac{(1 + \alpha \Delta v_x) + (\alpha \Delta v_x - 1) \tanh(-\Delta x \Delta v_x)}{2} \tau_x \quad (3)$$

$$\tau(\Delta y, \Delta v_y) = \frac{(1 + \alpha \Delta v_y) + (\alpha \Delta v_y - 1) \tanh(-\Delta y \Delta v_y)}{2} \tau_y \quad (4)$$

Here $\Delta x, \Delta y$ are the relative position wrt obstacle vehicle, $\Delta v_x, \Delta v_y$ the relative velocities wrt obstacle vehicle.

1.1 Potential Field Plots



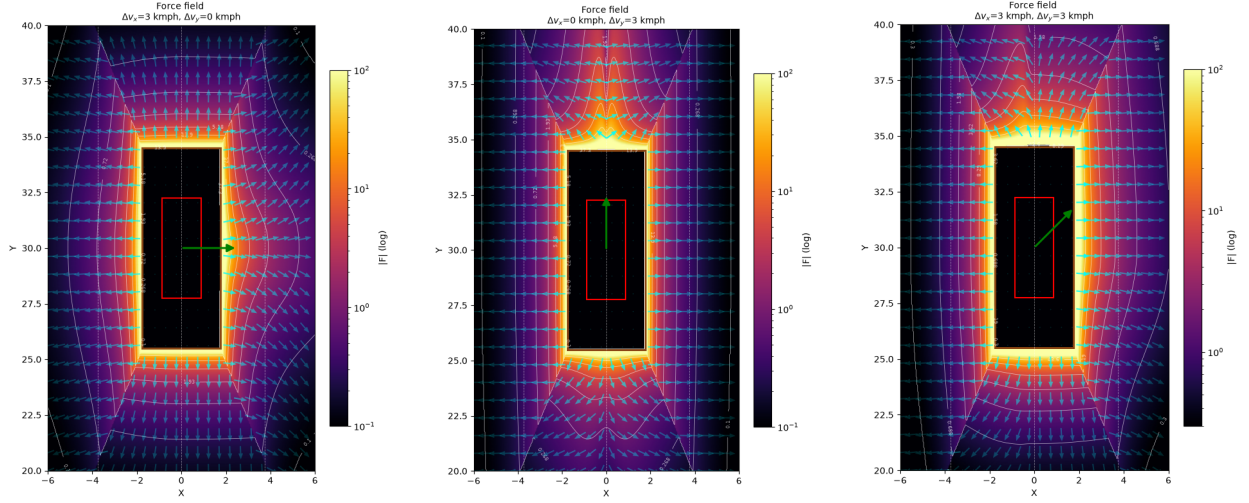
(a) Varying Δv_x

(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 1: Obstacle potential field U_{obst} .

1.2 Force Field



(a) Varying Δv_x

(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 2: Force field derived from U_{obst} .

1.3 Sensitivity to τ_x, τ_y

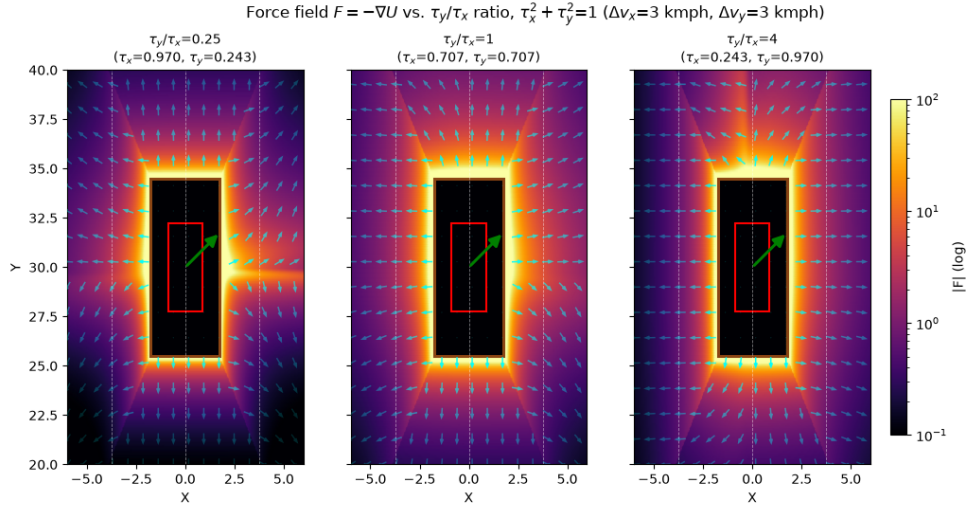


Figure 3: Sensitivity of the field to τ_x, τ_y .

1.4 Velocity at a Fixed Point

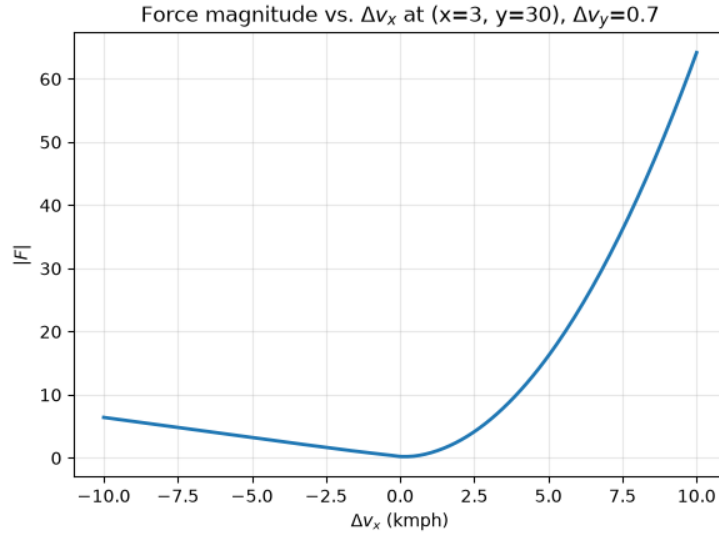


Figure 4: Velocity at a fixed point.

2 Possible Issues

2.1 Anomalous Trough in the Force Field

Figure 2b (varying Δv_y , Section 1.2) shows a localized trough in the force field rather than a smooth, monotonic falloff. This non-monotonic behavior suggests the force can momentarily weaken in a region where it should be increasing (or vice versa), which may translate into inconsistent or jerky avoidance behavior for the controlled vehicle as it crosses that region.

2.2 Sensitivity to τ_x, τ_y and Parallel Forces

As shown in Section 1.3, the shape of the potential (and hence the resulting force) is highly sensitive to the choice of τ_x and τ_y . Small changes in these parameters can noticeably reshape the field. Additionally, in some regions the resulting force vectors become nearly parallel to one another rather than radiating outward from the obstacle, which weakens the field’s ability to push the vehicle laterally away from the obstacle and can instead leave it drifting alongside it.

2.3 Fixed Velocity Behavior

The velocity profile at a fixed point (Section 1.4) stays essentially constant rather than responding to the surrounding potential. Since the force field is meant to drive velocity changes, a velocity that remains fixed at a given location raises the question of whether the coupling between the potential/force field and the vehicle’s velocity update is behaving as intended.

2.4 Overparameterization Relative to the Degrees of Freedom

The model introduces several free parameters ($\lambda, \alpha, \tau_x, \tau_y, w, l, \Delta v_{max}$) to shape a field that ultimately governs a comparatively low-dimensional system (position and velocity in two dimensions). Having more tunable parameters than underlying degrees of freedom makes the fit under-constrained: multiple, quite different parameter combinations can produce similar overall behavior, which complicates both calibration and the interpretability of any single fitted parameter.